

Acidic extraction of histone proteins

Risk assessment

- Work with highly flammable liquids
- Work with concentrated acids
- ▷ Handle acids and flammable liquids in a certified chemical fume hood
- ▷ Wear gloves, safety glasses, lab coat
- Collect acetone as ORGANIC SOLVENT WASTE



Reviewed: Oct 26, 2023



This is a bench card. Full protocol available online.

>> Acid extraction

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| ☐ 0.2 M Hydrochloric acid, 1 mL | ☐ 99.9% Acetone, 1 mL |
| ☐ 100% Trichloroacetic acid | ☐ 100% Acetone |

(1.) Start with a nuclei pellet or a nuclear fraction isolated from approximately 1.0×10^6 – 1.0×10^7 cells. Approximate the packed cell volume; this is 1 vol. 

(2.) Gently vortex the pellet from the preceding nuclear isolation or extraction procedure.

(3.) **Critical:** In a dropwise manner, add 5 vol (10 vol if starting with fewer than 1×10^6 cells) of 0.2 M hydrochloric acid to the pellet at 4 °C. Gently pipette up and down to disperse the pellet.

Critical: Add slowly to avoid rapid precipitation. If working with small volumes, pipette gently rather than dropwise. 

(4.) Incubate on ice for 5 min.  5 min

(5.) Centrifuge at $15\,000 \times g$ for 5 min at 4 °C. Transfer the supernatant to a clean tube. 

(6.) To the supernatant, add 0.25 vol of 100% trichloroacetic acid. Vortex vigorously to precipitate histones. 

Critical: Mix immediately and completely to avoid incomplete precipitation. 

(7.) Precipitate on ice for 5 min.  5 min

(8.) Collect the histone pellet at $15\,000 \times g$ for 5 min at 4 °C. Discard supernatant. 

(9.) Carefully add 100 µL cold (–20 °C) 99.9% acetone with HCl to the pellet. Do not disturb the pellet! 

Quality assurance: Since acetone can degrade some plastics, perform washes quickly in compatible tubes. 

(10.) Centrifuge at $15\,000 \times g$ for 2 min at 4 °C. Remove acetone using a glass pipette.

Quality assurance: Do not scrape the walls. Leave residual solvent rather than risk losing pellet material. A small amount of acetone will evaporate quickly. 

(11.) Repeat acetone wash using 100 µL pure acetone (without HCl).

(12.) Air dry the pellet at room temperature.

Critical: Avoid overdrying. Desiccated pellets may be difficult to redissolve. 

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Storage of isolated histones

- (1.) Store the histone pellet at room temperature for up to one week. For long-term storage, keep at -80°C . 

 Recipe (available online)  Notes (available online)

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